MIL-T-81546A(AS)
18 February 1971
Superseding
MIL-T-81546(AS)
21 July 1967

MILITARY SPECIFICATION

TEST STAND, ACTUATOR, LINEAR LOAD

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for a linear load, actuator, test stand (herein referenced test unit). This test unit is a system consisting of two upright, stationary, pneumatically powered test stands and two separate bench type control cabinets for use in performing operational tests on Linear Actuators.

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

W-C-596	Connector, Plug, Electrical, Connector Receptacle, Electrical; Plates, Wall Electrical
GG-G-76	Gauge, Pressure and Vacuum, Dial Indicating, For Air, Steam, Oil, Water, Ammonia, and Chloro-Fluoro Hydrocarbon Gases
PPP-B-601	Boxes, Wood, Cleated, Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connector Electric, AN Type

FSC-4920

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MIL-M-10304	Meters, Electrical, Indicating, Panel Type, Ruggedized
MIL-C-13924	Coating, Oxide, Black for Ferrous Metal

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with appropriate test methods

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 Screw Threads, Standards for Federal Services

Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D.C.

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C-1 National Electrical Code

Application for copies should be addressed to American National Standards Institute, 10 East 40th Street, New York, N.Y.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules

Application for copies should be addressed to the American Trucking Association, 1616P Street NW, Washington, D.C.

3. REQUIREMENTS

3.1 First article - The test unit furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to applicable specifications or standards and shall be as specified herein. Materials which are not covered by applicable specifications or standards shall be of the best quality, and entirely suitable for the purpose intended.

3.3.1 Metals - Metals shall be of the corrosion-resistant type or treated to resist corrosion.

3.3.2 Metal parts - All metal parts shall be of corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

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3.3.3 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Fungus proof materials - Materials that are nutrients of fungi shall not be used.

3.4 Design and construction - The test unit shall be designed and constructed to withstand the stresses, jars, vibration and other conditions incident to shipping storage, installation and service. The design and configuration of the test unit shall consist of two free standing load fixtures (main fixture and mini fixture) for positioning small and large size Linear Actuators and applying both tension and compression loads. The controls shall be contained in two separate bench type control cabinets, one for the pneumatic system and one for the electrical system. The cabinets are to house all controls, instruments, gages, switches, etc., as are necessary to conduct and monitor static and operational test.

3.4.1 Capacity - The main fixture shall be capable of handling actuators having an extended length of up to 52 inches which includes a maximum stroke of 30 inches. The mini fixture shall be capable of handling actuators having an extended length of up to 22 inches which includes a maximum stroke of 18 inches. Horizontal clearance between structural uprights must be a minimum of 14 inches for the main fixture, and a minimum of 11 inches for the mini fixture. The main fixture power head shall be capable of developing a minimum loading of 4500 pounds in compression and 4300 pounds in tension when supplied with compressed air to 100 PSI. The mini fixture power head shall be capable of developing a minimum loading of 400 pounds in compression and 350 pounds in tension when supplied with compressed air at 100 PSI. The mini fixture shall have a useable range of from 10 pounds to 400 pounds.

3.4.2 Threads - All threaded components shall conform to the applicable requirements of the National Bureau of Standards Handbook H-28.

3.5 Maintainability - The test unit shall be constructed to facilitate ease of operation, maintenance and repair.

3.6 Performance characteristics - The test unit and component parts shall meet the performance tests in Section 4 in the order listed below:

(a) Operational test (4.6.2)

(b) Static test (4.6.3)

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(c) Dielectric strength test	(4.6.4)
(d) Instrument test	(4.6.5)
(e) Timer test	(4.6.6)
(f) Switch test	(4.6.7)
(g) Radio interference	(4.6.8)

3.7 Details of components - The test unit shall consist of two load stands (main fixture and mini fixture) and a control console.

3.7.1 Load stands description - Both main fixture and mini fixture shall consist of a base, two (2) structural uprights, fixed upper cross beam, power head, movable carriage, anchor device (base anchor for mini fixture, and adjustable anchor beam for main fixture), and related components as are required to insure satisfactory positioning and loading of Linear Actuator.

3.7.1.1 Base - The main fixture and mini fixture shall have a solid steel base of such size and shape as to insure free standing stability. Each base is to contain sockets for the uprights, and pads at each corner. The pads shall be floor type for the main fixture, and bench type for the mini fixture. The pads shall be adjustable in the base and swivel to provide stability on irregular surface.

3.7.1.2 Structural uprights - The structural uprights shall serve as guide rails and as an inner bearing race for the guide bushing bearing balls. They shall be solid steel bar, hardened and ground within .0005 inches of nominal diameter with a finish of 16 micro inch RMS, straight within .001 inch per foot, and parallel from top to bottom to insure free operation of the movable carriage. The structural uprights for the main fixture only shall be drilled to permit vertical adjustment of anchor beams on 3 inch centers for five different positions.

3.7.1.3 Fixed upper cross beam - To be of steel fabricated so as to provide sockets for the structural uprights and serve as attachment point for the pneumatic power head.

3.7.1.4 Power head - Both main fixture and mini fixture shall consist of a double acting pneumatic cylinder, and fittings, of such size and shape as to permit development of full capacity as specified in paragraph 3.4.1.

3.7.1.5 Movable carriage - To be the means by which one end of the actuator is secured and through which the load is applied. To consist of a movable steel member captive to the structural uprights and guided at the outboard ends. The

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carriage shall have a threaded attachment with split end to permit securing rod end of the actuator and a coupling to permit securing of the pneumatic cylinder piston rod end.

3.7.1.5.1 Attachment pins and bushings - The split end of the threaded attachment shall have two threaded in-line holes fitted with removable bushings to permit insertions of load pins. Load pins must be centrally located in the bushings their purpose being to couple various size actuator rod ends. Two pins and four threaded bushings of the following sizes are to be furnished with each unit: .2495", .3120", .3745", .4375", .4995", and .5620" diameter for the main unit; .1245", .1557", .1870", and .2495" diameter for the mini unit, these dimensions are bushing bore diameters and the matching pins are to be ground for a snug sliding fit. The pins are to have a 16 micro inch RMS finish and a life expectancy of 1,000,000 reversals. A steel carrying case shall be provided to house all threaded bushings and attachment pins. The bushings shall be grouped as to size and identified by labeling in the kit.

3.7.1.5.2 Guide bushings - The outboard ends of the movable carriage are to be guided by precision linear ball bushings having multiple ball circulating tracks and having a coefficient of friction of .002 - .004.

3.7.1.6 Anchor device - To be of steel fabricated so as to provide a means for attaching the actuator. The main fixture only shall have an adjustable anchor beam of steel. A box-section adapter for mounting flange-type actuators capable of full-load capacity shall be provided for the main fixture. The adapter shall attach to the fixed beam. Beams or adapters to be removable from vertical uprights and adjustable within full range. Universal adapters are to be provided so that a wide variety of actuator mounting can be accommodated. For the mini fixture the universal adapters shall be attached directly to the load fixture base.

3.7.2 Control console -

3.7.2.1 Control cabinets, general - Cabinets are to be of sheet metal and such design and construction as to withstand severe service. Maximum dimensions are to be 24" wide (L to R) x 17" deep (F to B) x 15" high having a sloping front for instrument and control mounting. Cabinets are to be portable, enclosing all controls both pneumatic and electric and affording means for providing service to test fixtures and a means for monitoring actuator performance under test. All interconnections, both pneumatic and electrical, between cabinets and test fixture are to be by removable hose and conductors fitted with end fitting of such design as to insure quick disconnect and insure operator safety. All indicating dials and controls are to be mounted on the cabinet sloping front and to be so positioned as to be convenient for reading and operation from the operators normal position.

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All dials, gages, etc., are to be face mounted and top side access is to be provided for maintenance of components requiring frequent service. Operating switches, indicator lights, etc., may be mounted on straight portions of cabinet front and top. Recessed finger cavities are to be built-in to cabinet side to facilitate carrying.

3.7.2.2 Control cabinet, pneumatic - A metal cabinet detailed in 3.7.2.1 shall house all pneumatic control components, piping, gages, valves, etc., as are required to conduct test operations. Quick disconnect fittings, for incoming air and air supply to test fixtures, are to be from rear of cabinet. The system must be capable of operation on 100 psig compressed air supply. Means are to be provided for infinite regulation of delivered air over full pressure range. The system must be designed so as to permit venting air pressure over a regulated setting while applying an aiding or opposing load. A three position control valve must be provided to select tension, compression or neutral position. Regulating and control valves must have sufficient flow and pressure capacities for continuous duty on 100 psig air supply. Regulation of air pressure must be maintained within 2 psi of set pressure under either aiding or opposing loading. An air filter and air line lubricator are to be included in the pneumatic circuit and be accessible for maintenance and inspection. A dial indicating gage for both tension and compression shall be located on output side of pressure regulator. All gages, pressure regulator operating knob and selector knob are to be mounted in the sloping face of the control cabinet and be convenient for the operator.

3.7.2.2.1 Gages - All pneumatic gages must have full range capacity and a built-in accuracy of 1/2 of 1 percent. Gages are to be a minimum of 6 inches in diameter and must have an external adjustment for calibration. Gages for both tension and compression are to be face mounted within the cabinet sloping front. Gauge dials shall have color-coded scales to directly indicate load applied for either fixture over full range. Also, a psi scale shall be provided, 0-100 psi, for calibration. Gages shall conform to GG-G-76.

3.7.2.3 Control cabinet, electric - A metal cabinet detailed in 3.7.2.1 shall be provided to house all electric control components, timer, meters, switches, circuit breakers, terminals, etc., as are required to conduct test operations. Connections between control cabinet and actuator under test are to be portable cords having ends suitable for connecting to binding post at control cabinet and open end leads at actuator. Each cabinet to be furnished with a total of twelve (12) leads (nine (9) for A.C. and three (3) for D.C.) capable of delivering design capacity of system. Power for operation of the actuators is to be from external

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power sources, delivered to and monitored through the control cabinet. Input power connections are to be by portable cords, from rear of cabinet, receptacles and plug to be in accordance with W-C-596 and MIL-C-5015. Output power requirements are 110 Volt, single phase, 400 cycle, 10 amperes; 208 Volt, 3 phase, 400 cycle, 10 amperes; and 28 Volt, D.C., 20 amperes. Control circuit to operate on 115 Volt, single phase, 60 cycle service. Instrumentation to include A.C. and D.C. volt meters and ammeters with selector switch to permit reading of total draw in any phase, timer, operational switches for forward and reverse operation, indicator lights, terminal binding post, voltage and phase selector switches, and all other components required for safe and efficient operation. All gages, instruments, switches, terminals, circuit breakers, indicator lights, etc., are to be face mounted on cabinet front. Components not requiring visual readout may be mounted on straight portion of cabinet face or top. All instruments, switches, indicator lights, terminals, etc., are to be clearly labeled with rigid nameplates conforming to MIL-STD-130 and securely affixed in place or have the identification engraved or silk screened on the cabinet front panel in a relative position as applicable. The lettering is to be a minimum of 1/8" high; and if engraving is used, characters are to be filled with black enamel to insure visibility. All circuit breakers are to be of the reset type and to be available for reset without having to remove panels or other covers.

3.7.2.3.1 Power cable - The electrical control cabinet shall be furnished with a separate electrical supply cord for each required external power source. Each cord shall be 10 feet long conforming to ANSI-C-1. The power cord shall be fitted with applicable connector plugs in accordance with W-C-596. The size and type shall be suitable for the required service.

3.7.2.3.2 Power switch - A power ON-OFF switch shall be provided for both A.C. and D.C. sources. An indicator lamp shall be provided to indicate "POWER-ON".

3.7.2.3.3 Circuit breaker - All electrical circuits must be protected by circuit breakers of adequate capacity to permit operation at full range of components. All circuit breakers must be accessible from front of console.

3.7.2.3.4 Timer - Actual time of load test must be measured by means of an electric timer. Timer must be capable of automatic or manual operation or control by external switches. Timer to have a one second, 360° sweep with dial graduations in.01 second; a one minute (0-60 second) hand, 360° sweep with dial graduations in one second; and a 0-15 minute totalizer in one minute graduations to be at least 3.5 inches in diameter and face mounted within the sloping face of the cabinet. Timer control circuit is to incorporate a electronic, solid state current sensing device to automatically start and stop timer consistent with current flow to the actuator and to be capable of operation in both the A.C. and D.C. modes. The sensor circuit shall activate the timer at any current draw between 150 miliamps and 20 amps.

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Response time shall be less than one milisecond. The timer is to be provided with a portable remote cord, at least three feet long, with start, stop, and reset buttons.

3.7.2.3.5 Instrumentation - Voltmeters and Ammeters, 2 1/2" minimum diameter, shall be provided to monitor power consumption during test. Meter internal movements to be of ruggedized construction conforming to MIL-M-10304. Meters are to be face mounted on the sloping face of control cabinet. Phase selector switch is required for reading current flowing in any one branch of a three phase system. A.C. ammeters to be dual range having scale of 0-10 and 0-1.0 amperes. D.C. ammeter to be dual range having scales of 0-20 and 0-2.0 amperes. Range selector switch to be panel mounted with spring return momentary contact in low range position to prevent meter damage due to in-rush current at start-up.

3.7.2.3.6 Run-job switch - A three position switch shall be mounted on the panel to operate the item being tested. One position shall be spring-loaded for momentary contact to be used to jog the actuator during set-up and special testing.

3.7.2.3.7 Mode selector switch - A "Mode Selector Switch" shall be provided to select automatic, manual, or auxiliary modes of operation of Actuator being tested and inter-related circuitry. This switch shall be so wired as to allow operation of test actuator through timer remote cord.

3.7.3 Label plate - Each switch, gage, meter, indicator light, terminal, control device, etc., shall be labeled as to identify its function. Power outlets are to be identified as to characteristics. Label plates are to be securely affixed to cabinet, or the information may be engraved or silk screened on the front panel in letters no less than 1/8" high and filled with black enamel if engraving is used.

3.7.4 Accessories - One complete accessory set shall be installed on each test fixture. The accessory equipment shall consist of a scale indicator and an adjustable microswitch. This accessory equipment will permit accurate adjustment of mid-position and verification of full travel capability. It will allow simplified calculations wherein actuator speed must be expressed in inches per second. The precision steel scale shall be mounted on one side of the test fixture. The scale shall be adjustable (to permit setting the "zero" reference point), and shall be equipped with two adjustable reference stops. A pointer shall be attached to the movable crosshead to allow the "stroke" to be read directly in inches. The adjustable microswitch shall be attached to the scale. The microswitch shall be furnished with leads and plugs to mate with the "auxiliary" terminals on the

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electrical console, and shall be actuated by the pointer on the movable crosshead. When in the "auxiliary" mode, circuitry within the console shall automatically stop the actuator and record time for any pre-determined intermediate actuator setting. Accessory equipment shall be mounted on the linear actuator test stand.

3.8 Color and finish - Load Fixture: All functional steel parts shall be finished black oxide in accordance with MIL-C-13924, Class 1. Non-functional parts may be painted grey enamel. Control Cabinet: All parts fabricated from ferrous metal not otherwise plated or protected, shall be painted grey enamel.

3.9 Identification marking - A metal nameplate of reasonable size, permanently marked in accordance with MIL-STD-130 shall be securely attached to each of the test stands and the control cabinets.

3.10 Workmanship - Workmanship shall be in accordance with the best commercial standards. All components and assemblies shall be free from burrs, rough surfaces, marred finish, broken, loose damaged or missing parts; and be capable of functioning satisfactorily when properly connected to an adequate power source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure the supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of the main test fixture, mini test fixture and console shall be classified as follows:

(a) First article inspection (4.3)

(b) Quality conformance inspection (4.4)

4.3 First article inspection - First article inspection shall consist of all the inspections of this specification.

4.3.1 Sampling for first article inspection - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish one test unit for inspection to determine conformance with this specification. The sample shall be identified with the manufacturer's part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2).

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4.4 Quality conformance - Each test unit on the contract or order shall be subjected to the following inspections in the order listed. Any test unit containing a defect or failing to pass any of the inspections shall be rejected.

(a)	Examination	(4.6.1)
(b)	Operational test (except that 5 complete cycles will be executed)	(4.6.2)
(c)	Examination of preservation, packaging, packing and marking	(4.6.9)

4.5 Test conditions - Unless otherwise specified, tests shall be conducted at an ambient temperature of 77° ± 18° F and an ambient barometric pressure of 28 to 32 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied.

4.6 Inspection methods -

4.6.1 Examination - The main test fixture, mini test fixture, and control cabinets shall be thoroughly examined for design, construction, materials, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.6.2 Operational test - With the mini test fixture and control cabinets connected operate all components through their normal operating range. Continue operations until 100 complete cycles have been executed. A cycle will be considered as one complete stroke (advance and return) of the air cylinder. Observe all facets of operation making certain there is no binding, misalignment, blockage, leakage or other faults which will impair satisfactory operation. Repeat this test using the main test fixture.

4.6.3 Static test - With the main load fixture movable carriage extended to a 15 inch length and the adjustable anchor beam positioned approximately 12 inches from the base, connect the two split end couplings with a solid connector. Alternately apply 150 PSI air pressure to each end of the power cylinder. Complete 100 cycles of at least 30 seconds duration to each end of the power cylinder. Examination of exposed components shall show no evidence of deformation, leaks, or component failures. With the mini fixture movable carriage extended to a 9-inch length, connect the two split end couplings with a solid connector and repeat the above test.

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4.6.4 Dielectric strength test - All circuits shall be subjected to a dielectric test of twice the maximum voltage (RMS) of the circuit plus 1000 volts (RMS), between insulated circuits and ground. All direct resistance ground connections shall be disconnected during the test. Voltage shall be applied for a period of 60 seconds and shall gradually be raised to its full value.

4.6.5 Instrument test - All electric meters shall be submitted to a 500 cycles of zero to full scale deflection. Each cycle shall consist of a 5 second "on" period and 1 second "off" period. After test, meters shall demonstrate accuracy in accordance with 3.7.2.2.1.

4.6.6 Timer test - Electric timer shall be operated over full range for 500 cycles without evidence of failure. For this test a cycle shall consist of one period of operation over full scale, 0-60 seconds, and a manual reset back to 0. At end of this test the timer shall operate over its full totalize range, 15 minutes, against a clock of known accuracy and show no appreciable loss or gain.

4.6.7 Switch test - Switches shall be operated at full load for 500 complete ON-OFF cycles. Upon completion of test switches shall show no evidence of arcing or burning.

4.6.8 Radio interference - The electric circuitry shall be tested to demonstrate compliance with MIL-STD-461, Class IIB.

4.6.9 Examination of preservation, packaging, packing and marking - Preservation, packaging, packing and marking shall be examined to determine conformance with the requirements of Section 5.

4.6.10 Re-inspection and re-test - Rejected units shall not be submitted for re-inspection or re-test without the contractor first furnishing full particulars concerning previous rejections and measures taken to correct the defects.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level A - The load test stands, control console, and accessories shall be cleaned in accordance with procedures C-1 and preserved and packaged in accordance with method IIa of MIL-P-116.

5.1.2 Level C - The load test stands, control console and accessories shall be preserved and packaged in accordance with the suppliers standard practice.

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5.2 Packing -

5.2.1 Level A - The load test stands, control console and accessories packaged in accordance with 5.1.1 shall be overpacked in an overseas type shipping container in conformance with PPP-B-601 or PPP-B-621.

5.2.1.1 Blocking and bracing - Cushioning, blocking and bracing shall be in accordance with MIL-STD-1186.

5.2.2 Level C - The load test stands, control console and accessories shall be packed in shipping containers complying with Uniform Freight Classification Rules or with the National Motor Freight Classification Rules or with other carrier rules applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required in the contract or purchase order, marking on the packages and containers shall be in accordance with Specification MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use - The test unit covered by this specification is intended for the testing of electrical actuators, under simulated operating conditions, having operating characteristics within the capacity range of the test unit.

6.2 Information for contracting officer - Contracts or orders should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

(c) Special marking, if required (see 5.3).

(d) Items of data required (see 6.3).

(f) Engineering change procedure in accordance with NAVAIR INST 4275.3A. The Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360 (Attn: AIR-534), will be responsible for review of Class 1 changes under the contract. The Commanding Officer, Naval Air Engineering Center, Philadelphia, Pa. 19112 (Attn: GSED) will be responsible for approval of Class II changes.

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6.2.1 Contracts or orders should specify the following provisions for first article inspection.

6.2.1.1 Whether first article inspection is required. When a contractor is in continuous production of the test unit from contract to contract, consideration should be given to waive the first article inspections. If first article inspection is required, indicate:

(a) Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - For the information of the Contractors and Contracting Officers, any of the data specified in (a) subparagraph below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Instruction manual - The instruction manual shall include instructions completely describing the operation of the test unit, descriptive drawings, and complete maintenance instructions.

Preparing Activity
Navy - AS
(DoD Project No. 4920-N280)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-81546A(AS) TEST STAND, ACTUATOR, LINEAR LOAD

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 62 1426

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